Happiest Health Summit 2026 | Phase 1: Prototype and Code

GlucoTwin AI

A predictive digital twin that fuses a patient's health record with wearable glucose data to warn of a dangerous low up to 2 hours ahead, and explains why.

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A normal reading can hide a low that is coming

Now: 105 mg/dL

The app shows a normal value. Nothing looks wrong.

Next 30 to 60 minutes

The trend is falling fast. A recent meal, insulin and activity add downward pressure.

An hour later: below 70

Shakiness, confusion, falls, in severe cases seizures. Nobody was warned.

Illustrative example. Today's tools mostly show the current reading, or alarm after a threshold is crossed. They rarely say what happens next, or why.

Our question: can we warn early, explain the reason, and show how sure we are, so the warning can be trusted?

Who needs this, and why it matters in India

~101 million

people with diabetes in India (ICMR-INDIAB 2023). Approximate figure; verify before citing.

Patients

Insulin or sulfonylurea users, older adults, people living alone. Night-time and post-meal lows go unnoticed.

Doctors

See snapshots between visits. Need explainable decision support, not a black box.

Health system

Severe lows cause emergency visits and admissions. Primary centres with limited specialist access need remote, early, low-cost monitoring.

Collect, understand, predict, explain, act

1

Collect

EHR profile plus wearable glucose, meals, insulin, activity

2

Understand

Fuse both streams into one patient state

3

Predict

Glucose at +30 / 60 / 120 min and chance of a low

4

Explain

SHAP reasons in plain language, with uncertainty

5

Act

Clinician reviews and tests what-ifs, then decides

Decision support only. GlucoTwin never prescribes insulin doses; the clinician makes every treatment decision.

Meeting the brief: two data streams, fused

Static stream: simulated EHR

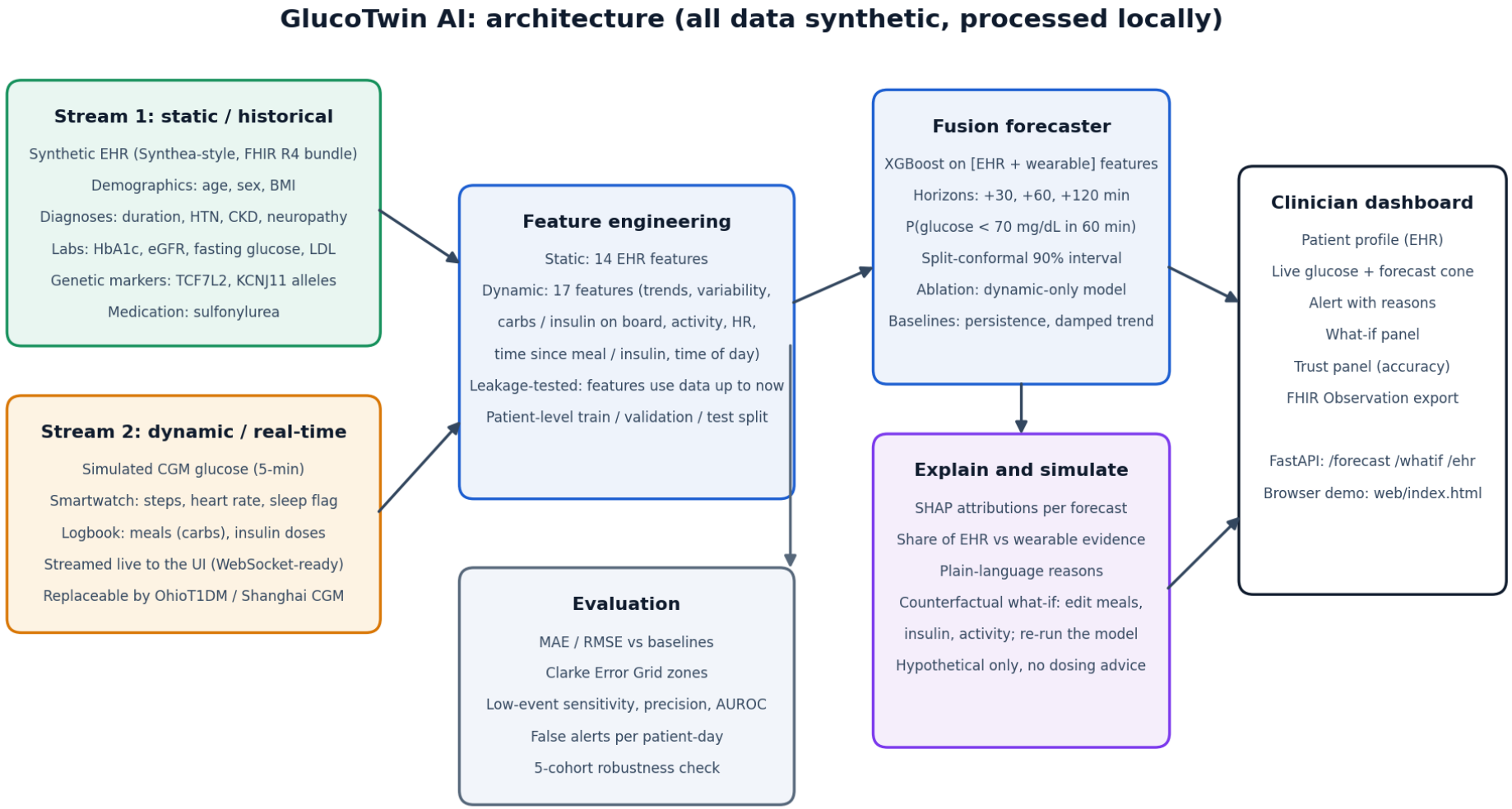
- Demographics: age, sex, BMI
- Diagnoses: duration, hypertension, neuropathy, kidney stage
- Labs: HbA1c, eGFR, fasting glucose, LDL
- Genetic markers: TCF7L2, KCNJ11 risk alleles
- Medication: sulfonylurea; FHIR R4-style export

Dynamic stream: simulated wearables

- CGM glucose every 5 minutes
- Smartwatch: steps and heart rate, sleep flag
- Logbook: meals (carbs) and insulin doses
- Streams live into the dashboard
- Swappable for open CGM datasets

Fusion: 14 EHR features + 17 wearable features go into one model, tested on 16 patients it never saw during training.

System architecture



Privacy by design: synthetic data only, no identifiers, local inference, no external AI API calls. Research prototype, not a medical device.

Data and privacy: built for the sandbox rules

Need	What we used	Status
EHR	Synthetic Type 2 diabetes cohort (60 patients), FHIR R4-style bundles	Generated
Sensor data	Simulated CGM, steps, heart rate, meals, insulin	Generated
Real datasets	OhioT1DM, Shanghai CGM, Synthea, MIMIC-IV: licences to confirm	Planned

- No real patient data and no identifiers anywhere
- Models and explanations run locally; no data sent to external AI APIs
- Designed in the spirit of DPDP Act 2023 and HIPAA-style limits (not a compliance certification)
- Caution: OhioT1DM is Type 1 data and must not be presented as Type 2 validation

The AI model and how we tested it

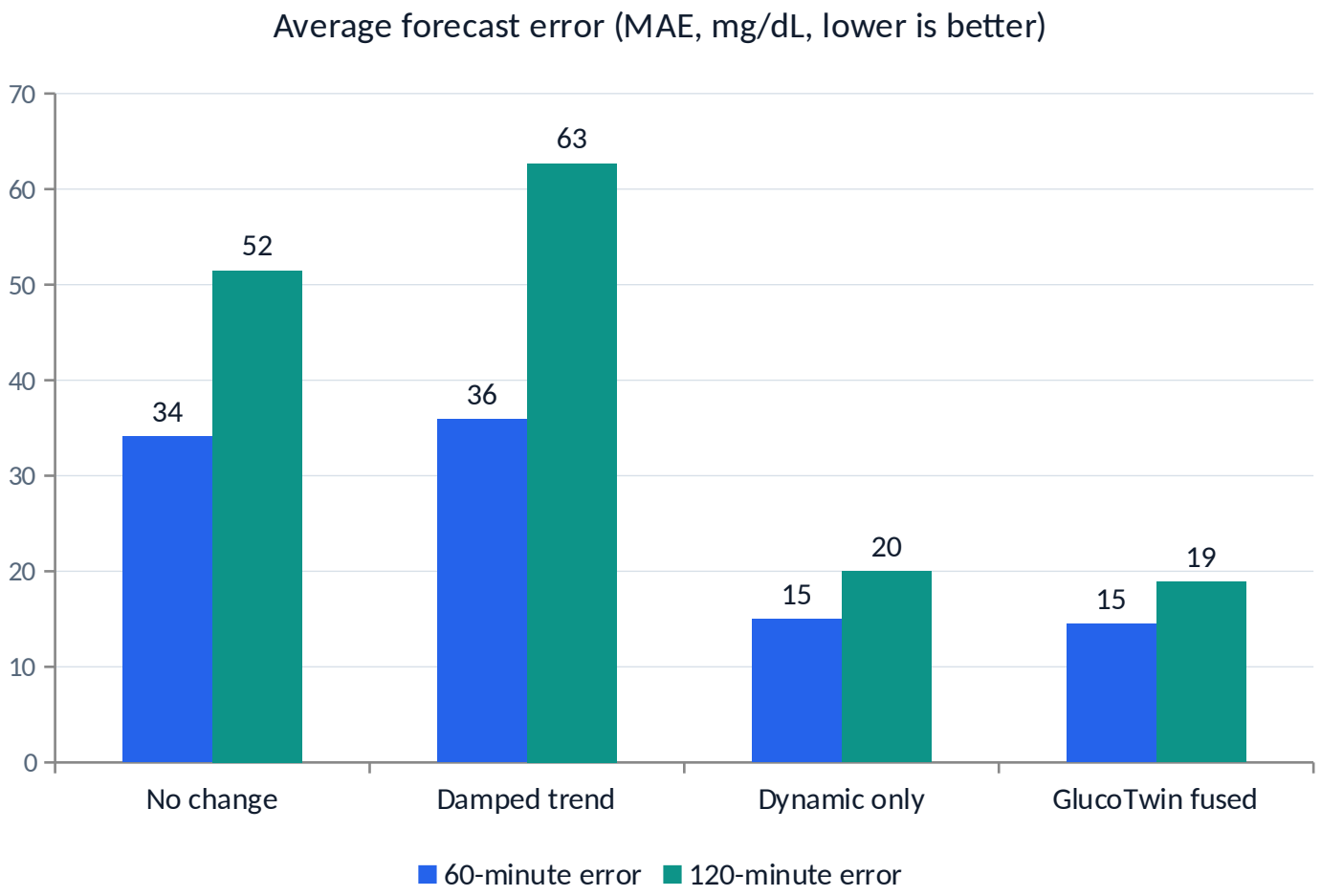
Model

- XGBoost forecaster per horizon: +30, +60, +120 minutes
- Classifier: chance of glucose below 70 mg/dL in 60 minutes
- Inputs: fused EHR + wearable features (trends, carbs and insulin on board, activity, heart rate, time of day)
- Split-conformal 90% intervals show uncertainty
- SHAP explains each forecast; what-if re-runs the model

Honest testing

- Patient-level split: 36 train, 8 validation, 16 unseen test patients
- Features use data up to now only; a unit test checks for leakage
- Baselines: no change, damped linear trend
- Ablation: same model without EHR data
- Metrics: MAE, RMSE, Clarke Error Grid, sensitivity, precision, AUROC

Results on 16 unseen synthetic patients



14 vs 34

mg/dL average error at 60 min, fused vs no change

0.97

AUROC for spotting a low in 60 min

90%

of outcomes fell inside the 90% interval

What the results do and do not show

What works

- Beats simple baselines clearly
- Intervals are well calibrated (about 90%)
- Reasons are explainable per forecast
- Pipeline is reproducible and tested (7 tests)

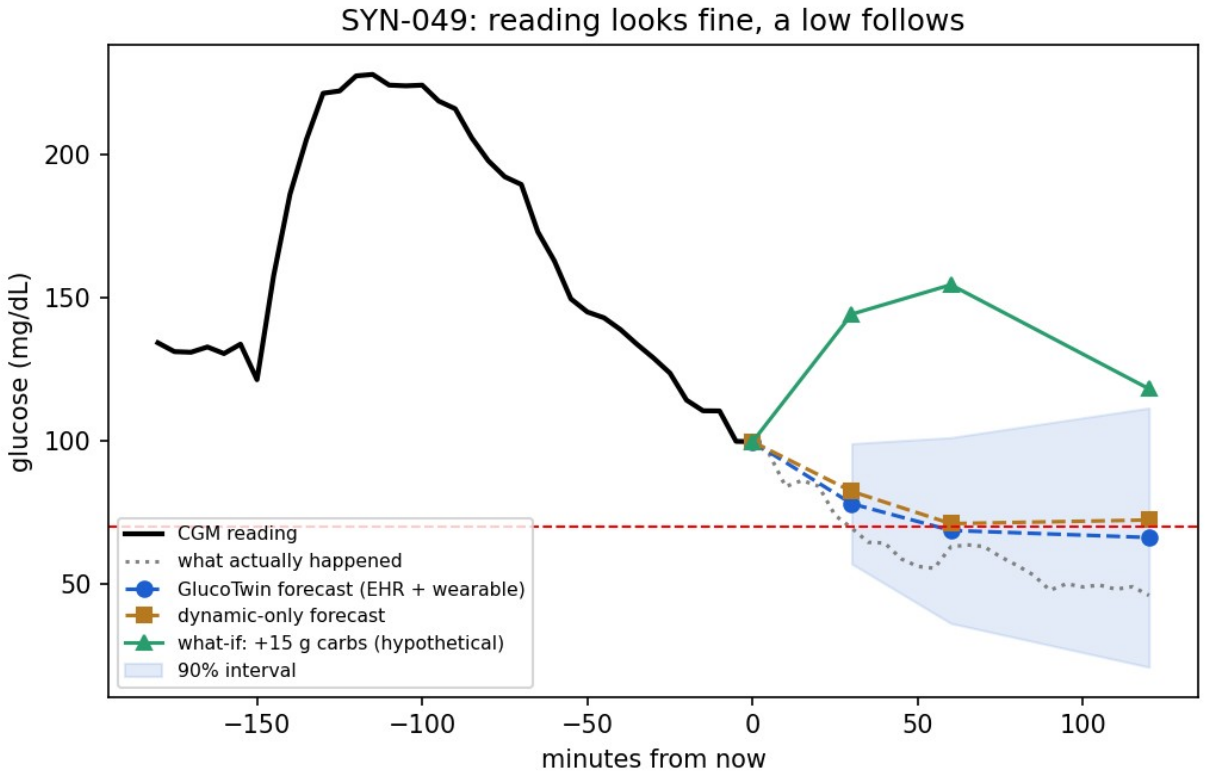
What is modest

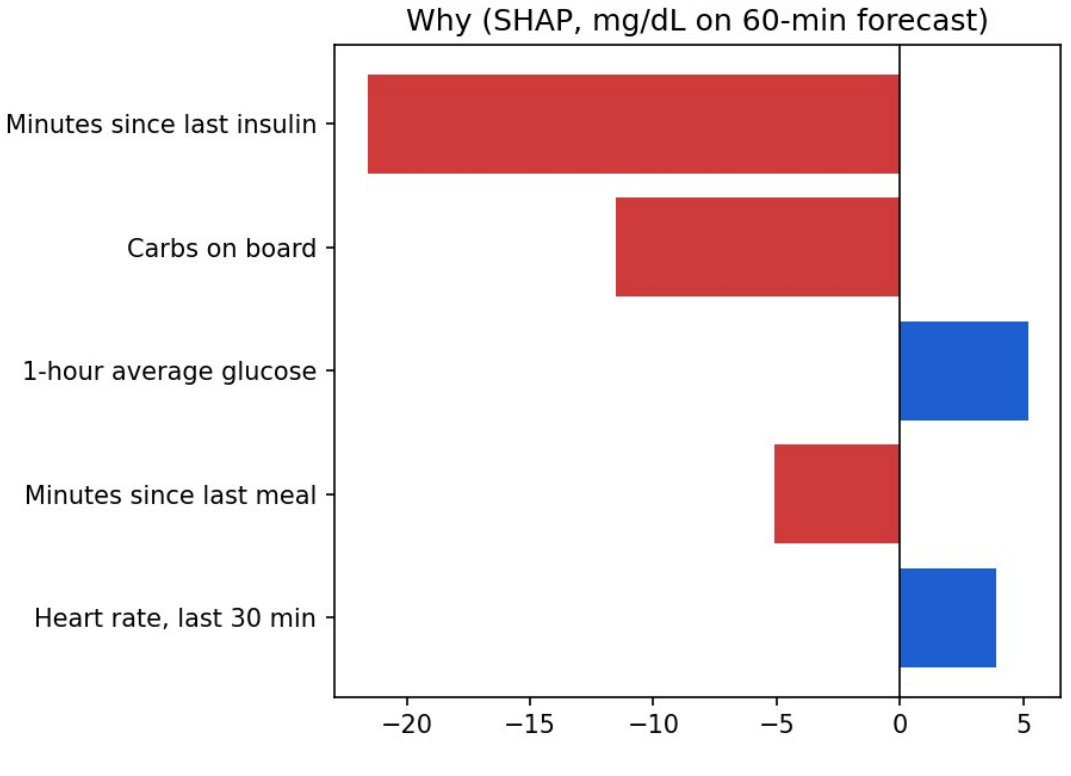
- EHR fusion gain is small: about 0.6 mg/dL at 60 min, better in 5 of 5 cohorts
- EHR carries 9% of SHAP importance
- Alert rule: 74% sensitivity, 70% precision, about 9 false-alert readings per patient-day

Not proven

- Clinical validity: all data is synthetic
- Gain depends on assumed EHR-physiology links
- Real CGM data is noisier
- No patient or clinician study yet

Demo: a normal reading, a low is coming





Held-out patient SYN-049: glucose 100 mg/dL now, actual 63 an hour later. Fused forecast 69 warns; the dynamic-only model says 71 and stays quiet. Bars show why. The green line is a labelled what-if: 15 g of carbs.

The clinician and patient dashboard

1. Play the live stream

Readings arrive every 5 minutes. The gauge, trend and forecast cone update live.

2. Read the warning

Plain-language headline, time to predicted low, and the 1-hour range.

3. See what is driving it

Reasons such as insulin still active or recent food, with mg/dL impact.

4. Try a what-if

Snack, meal, half the active insulin, or a walk. The model recalculates; labelled hypothetical.

Also: load your own CSV, see time in range today, a trust panel with accuracy, FHIR export, and a REST API.

Responsible by design

R1 Explain every alert

SHAP drivers and plain-language text on each forecast.

R2 Show uncertainty

Calibrated 90% interval; wide band means low confidence.

R3 Never advise doses

What-if is hypothetical and says so. Clinician decides.

R4 Work without real data

Synthetic EHR and sensors; local processing.

Impact and roadmap

Readiness: TRL 4, an integrated prototype tested on synthetic data in a lab setting

Next 3 months

- Train on open CGM data (licences first)
- LSTM / Temporal Fusion Transformer with cross-attention
- Reduce false alerts; per-patient thresholds

6 to 12 months

- Retrospective validation on authorised clinical data
- Clinician usability study with diabetologists
- Remote-monitoring pilot for primary centres

Before any patient use

- Regulatory review as software as a medical device
- Security and consent audit
- Prospective clinical evaluation

Thank you

GlucoTwin AI: predict, explain, and let the doctor decide.

- Code, README, architecture and results: github.com/AnnuAnshu26/Powerpuff-IGDTUW
- Open-source licence: MIT
- Team Powerpuff Girls: Anushka Aggarwal, Anshu Yadav, Aarti | IGDTUW